# The Reconstructed Theory Has One Mass: a Machine-Checked Volume-Uniform Spectral Gap for a Concrete Transfer Operator

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DRAFT — not for distribution; live slots present

**Abstract**

For the spatial $\mathbb{Z}_2$ (Ising-slice) system in the Dobrushin window $2\tanh|\beta| + 2\tanh|\gamma| \leq \alpha < 1$, we machine-check in Lean 4 that the Osterwalder–Schrader (site-form) reconstruction of the transfer operator carries *one* mass: there is a single $m > 0$ such that for *every* spatial extent $L$ the reconstructed (site-form) transfer operator is unitarily conjugate — by the explicit $\sqrt{w}$ boundary dressing — to the symmetrised kernel whose projected norm the Dobrushin corollary bounds by $e^{-m}$, and every connected correlator of the transported transfer operator decays as $|\mathrm{connCorr}\, T\, \Omega\, v\, n| \leq \|v\|^2(e^{-m})^n$ — constant per-observable, rate alone uniform in the volume. All quantifiers live in the Lean statements; the prose here copies the kernel's cut. To situate the claim in the 2026 landscape of machine-checked constructive field theory: the abstract construction (reflection-positive data $\to$ physical Hilbert space $\to$ self-adjoint contractive transfer operator) has been machine-checked, sorry-free, in the Harvard formalization programme, *with the spectral gap carried as an undischarged hypothesis* (`GappedTransfer`); the Osterwalder–Schrader axioms have been machine-checked for the free field; and measure-side Dobrushin clustering has been machine-checked for lattice Yang–Mills. What we add, and what to our knowledge no machine-checked artefact yet contains, is the combination {concrete interacting model, gap *proved* rather than hypothesised, uniform in the spatial volume} on the reconstructed operator — i.e. a concrete discharge of exactly the hypothesis shape the abstract frameworks carry. We do not claim Wightman reconstruction, a continuum limit, $SU(N)$, or any progress on the Clay problem.

## 1 Introduction and landscape

The Osterwalder–Schrader route builds a physical theory out of Euclidean data: a reflection-positive form gives a Hilbert space, and composition with the time shift gives a self-adjoint transfer operator. For a lattice slice system the construction is elementary — the content is never the construction but the *spectrum*: a spectral gap at every fixed spatial extent $L$ is classical (Perron–Frobenius; in this programme it is the earlier `coupled_gap_all_eigenvalues`), and it degenerates as $L \to \infty$ unless something uniform survives. What a thermodynamic limit can actually consume is *one* $m > 0$ valid for every $L$ at once.

*With machine-checked artefacts produced on the programme's Colab plane; toolchain `leanprover/lean4:v4.29.0-rc6`, Mathlib pinned to `0764272048...`. Repository: THE-ERIKSSON-PROGRAMME, branch `d3-closure`, anchor `8127e5b52d75116b19c172ea022e12c1cab0f8c2`.

That uniform input exists in this repository: the Dobrushin corollary `dobrushin_ising_uniform_gap` produces, inside the window $2\tanh|\beta| + 2\tanh|\gamma| \le \alpha < 1$, a single $m > 0$ bounding the projected transfer operator of the tilted kernel at every extent. The present paper's contribution is the *identification*: the operator that bound governs *is* — unitarily, by the explicit $\sqrt{w}$ boundary dressing, with no spectral-transport theory — the transfer operator forced on the reconstructed (site-form) theory, and the bound transports to volume-uniform clustering of its connected correlators. The theorem is deliberately cheap: the same witnesses, no new estimate. Cheapness is the honesty of the claim — the uniformity is real because it is D-6's, and the reconstruction adds exactly one commuting square, machine-checked.

### 1.1 Machine-checked prior work, cited before a referee does

The following table is load-bearing and its rows were collected by a four-modality search on 2026-08-04 (evidence: `docs/OS-R-PRIORITY-COLLATION.md` in the repository).

| Artefact | What is machine-checked | What is not |
|---|---|---|
| `mrdouglasny/reflection-positivity` (sorry-free ∼2026-06) | RP theta $\to$ Hilbert space (completion/quotient) $\to$ self-adjoint contraction $T$; trace formulas; susceptibility bounds from a gap | the gap itself: carried as the `GappedTransfer` *hypothesis*; no spatial-volume-uniform statement; concrete instance is finite-lattice RP only |
| `OSforGFF` (arXiv:2603.15770) | all five GJ/OS axioms for the $d$=4 Gaussian free field, 0 sorries | reconstruction (listed as next milestone); any interacting model |
| `xiyin137/OSreconstruction` | Wightman-side GNS lane of the full OS$\leftrightarrow$Wightman theorem | the theorem: 53 sorries + 12 axioms at last README snapshot |
| `mrdouglasny/lgt` (complete 2026-05-04) | Chatterjee-style Dobrushin uniqueness for lattice YM, $U(n)$, with exponential clustering whose constants depend only on $(n, d, \beta)$ | an operator gap; any reconstruction; the statement is per-volume on the measure side |
| `pphi2` | transfer-matrix spectral machinery for $\varphi^4_2$ | rests on 27–29 project axioms |
| This paper | the $\sqrt{w}$-dressing unitary; the transported volume-uniform gap; volume-uniform clustering of the reconstructed operator (`os_reconstruction_uniform_gap`, `os_reconstruction_uniform_clustering`; core 8481, oracle 3017, 0 sorry) | Wightman axioms; continuum limit; $SU(N)$; Clay; the raw-measure $n$-step identification (named next step) |

## 2 The two geometries and the forced operator

Everything in this section is prior work of the repository, cited by module (the public trace of this lane is the ai.viXra `2607.009x` cluster); no new mathematics appears here. The slice system carries two pairings on complex boundary vectors: the *site* form, which weighs by $1/w$ (`siteForm`, module `SpatialReconstruction`), and the *bond* form carrying the decoupled kernel spatialKernel $\beta$ (`bondForm`). The transfer operator is *forced*: $T = D_w K$ is the unique operator with $\mathrm{site}(u, Tv) = \mathrm{bond}(u, v)$ (`transferOp_unique`), self-adjoint for the site form at every $\beta$ (`transferOp_selfAdjoint`) and positive for $\beta \geq 0$; reflection positivity of the underlying measure is the lane's earlier bond/site pair (modules `SpatialReflection`, `SpatialOS`). On the Dobrushin side, the same repository carries `symWeighted` $= \sqrt{w}K\sqrt{w}$ (module `PerronGap`), its normalisation `tiltKernel` $=$ symWeighted$/\lambda$ (module `DobrushinTilt`), the Euclidean packaging `opOf`, `vacOf`, `vacuumTransfer_opOf` (module `DobrushinTransport`), and the corollary `dobrushin_ising_uniform_gap` (module `DobrushinCorollary`). The seam fact this paper depends on — one `spatialKernel` declaration serving both geometries — is enforced by the elaborator in the module of Section 3.

## 3 The dressing unitary

All three statements are green on the plane (module `YangMills/OS/OSReconstructionUniform.lean`, sha256 `5445429f...`, five-pass ladder, `ELAB_EXIT 0`). Write $Qu := \sqrt{w}\,u$ for the boundary dressing of a real Euclidean vector (`qEmbed`); the site form weighs by $1/w$.

**Theorem 3.1** (A1, `siteForm_qEmbed`). *For $w > 0$ and real $u, v$:* $\mathrm{siteForm}\, w\,(Qu)\,(Qv) = \sum_\sigma u_\sigma v_\sigma$. *Per site the cancellation is elementary:* $\sqrt{w}u \cdot \sqrt{w}v \cdot (1/w) = uv$.

**Theorem 3.2** (A2, `transferOp_qEmbed`). $T(Qu) = Q\big(\mathrm{act}\,(\mathrm{symWeighted}\, w\, \beta)\, u\big)$ *pointwise: the forced transfer operator on a dressed vector is the dressing of the symmetrised kernel's action.*

**Theorem 3.3** (A3, `transferOp_iterate_qEmbed`). $T^{[n]}(Qu) = Q\big((\mathrm{act}\, S)^{[n]}u\big)$ *— the shape the in-tree measure bridge `gibbsPathSum_eq_iterate` consumes.*

## 4 The transported volume-uniform gap

**Theorem 4.1** (`os_reconstruction_uniform_gap`). *In the window* $2\tanh|\beta| + 2\tanh|\gamma| \leq \alpha < 1$ *there is $m > 0$ such that for every $L$ there are $\lambda > 0$ and $\Omega > 0$ with the fixed-point equation* $\sum_\tau \mathrm{tiltKernel}\,(\mathrm{sliceW}\,\gamma\, L)\,\beta\,\lambda\,\sigma\,\tau\,\Omega_\tau = \Omega_\sigma$*, the projected bound* $\|\mathrm{projectedTransfer}\ldots\| \leq e^{-m}$*, and the A2 identity at* $w := \mathrm{sliceW}\,\gamma\, L$.

The proof is an `obtain` on `dobrushin_ising_uniform_gap` followed by A2: *the same witnesses*, no new analytic content — which is precisely the point. The uniformity is real because it is D-6's; the reconstruction contributes the identification, not a new estimate.

## 5 Identification against the measure

Green at pass 5 (operator side):

**Theorem 5.1** (B0b, `act_symWeighted_eq_smul_act_tilt`). $\mathrm{act}\, S\, u = \lambda \cdot \mathrm{act}\,(\mathrm{tilt})\, u$ *for* $\lambda \neq 0$.

**Theorem 5.2** (B0a, `act_iterate_eq_opOf_pow`). $(\mathrm{act}\, M)^{[n]}u = (\mathrm{opOf}\, M)^n(\mathrm{toLp}\, u)$ *pointwise.*

**Theorem 5.3** (`os_reconstruction_uniform_clustering`). *In the window: ONE $m > 0$ such that for EVERY spatial extent $L$, every connected correlator of the transported transfer operator obeys* $|\mathrm{connCorr}\, T\, \Omega\, v\, n| \le \|v\|^2 \cdot (e^{-m})^n$. *The constant is per-observable; the rate alone is uniform in $L$.*

The packaging is by citation: `vacuumTransfer_opOf` + `tiltKernel_symm` + `clustering_of_gap`, all in-tree. One step of the campaign charter remains open, and we name it rather than blur it: the *raw-measure* $n$-step identification — the statement that the unnormalised Gibbs two-point sums (`gibbsPathSum`, whose operator bridge `gibbsPathSum_eq_iterate` is already in the tree) equal $\lambda^n$ times matrix elements of the operator powers through B0a/B0b and A3, together with the division-safe six-term connected bound against the partition function. Its bricks are designed and its judge identities (G5 of `judge_os_uniform.py`) are verified numerically at $n \le 4$; it is fabrication, not research. Nothing in Sections 3–5 depends on it: the clustering theorem above is a statement about the reconstructed operator and stands on its own.

# 6 Reproduction

All numbers below were read from committed run artefacts (ledger Addenda 616–618); predictions were committed *before* their measurements.

| Quantity | Predicted | Measured |
|---|---|---|
| core jobs, unwired | 8480 (ledger baseline) | 8480 |
| core jobs, wired | $8480 + 1$ | 8481 |
| oracle reports, wired | $3010 + 7$ | 3017 |
| `sorryAx` occurrences | 0 | 0 |
| OS-R endpoint axiom sets | standard triple | exactly `[propext, Classical.choice, Quot.sound]` $\times 7$ |

Module: `YangMills/OS/OSReconstructionUniform.lean`, sha256 `5445429f7736...30cad1` (12,347 bytes), five elaboration passes on a fresh high-RAM Colab runtime (toolchain `v4.29.0-rc6`, Mathlib pinned); wired at anchor `8127e5b52d75116b19c172ea022e12c1cab0f8c2`. The count prediction $+1/+7$ held exactly — the ninth consecutive exact count prediction of the programme's D/OS lane. **[SLOT: Fresh-clone second witness at the anchor: judges both modes, core total, oracle report count with a continuation-joining axiom checker, and the sha256 list — from the pass-7 run output.]**

# 7 What is not claimed

No Wightman axioms, no OS→Wightman analytic continuation, no continuum limit, no $SU(N)$, no statement about the Clay problem. The window is the Dobrushin window; outside it the programme's own kill-test (repository, Addendum 610) shows the standard-cone Birkhoff route is projectively blind to the site weight, which is why no claim beyond the window appears here.